Experiment No.: 05

Title: Analyze random variables using probability distribution

Batch: B3 **Roll No.:** 16014223083 **Experiment No.:** 5

Aim: To analyze random variables using probability distribution

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Resources needed: VSCode, Python(NumPy, Pandas, Scikit-learn), Google Colab, Jupyter Notebook

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Pre Requisite: Data Science, Machine Learning

Theory:

A **random variable** is a numerical outcome of a random experiment. Random variables are classified into two main types:

- **Discrete random variables**, which take countable values (e.g., number of heads in coin tosses).
- **Continuous random variables**, which take values over a continuous range (e.g., time, height, or weight).

A **probability distribution** describes how probabilities are assigned to the values of a random variable.

- **Probability Mass Function (PMF):** Used for discrete random variables.

- **Probability Density Function (PDF):** Used for continuous random variables.

- **Cumulative Distribution Function (CDF):** Gives the probability that a random variable is less than or equal to a given value.

Algorithm / Procedure

1. Identify and define the associated random variable
2. Determine whether the random variable is discrete or continuous
3. Construct the appropriate probability distribution (PMF or PDF)
4. Verify that the total probability equals 1
5. Calculate the expected value (mean) of the random variable
6. Compute the variance and standard deviation
7. Plot the probability distribution
8. Interpret the results obtained from the distribution

Data Set

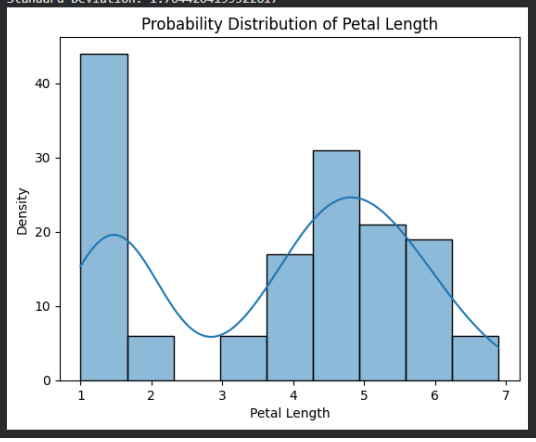
Iris Dataset/ Sales Dataset/ Savings Dataset

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load Dataset
df = pd.read_csv("Iris.csv")

# Step 2: Define Random Variable
X = df["PetalLengthCm"]

# Step 3: Identify Type
print("Random Variable Type: Continuous")

# Step 4: Verify Probability Concept
print("Using PDF (Histogram Density Approximation)")

# Step 5: Expected Value
print("Mean:", X.mean())

# Step 6: Variance and Standard Deviation
print("Variance:", X.var())
print("Standard Deviation:", X.std())

# Step 7: Plot Probability Distribution
sns.histplot(X, kde=True)
plt.title("Probability Distribution of Petal Length")
plt.xlabel("Petal Length")
plt.ylabel("Density")
plt.show()
```

Output:

```
Random Variable Type: Continuous
Using PDF (Histogram Density Approximation)
Mean: 3.758666666666666
Variance: 3.1131794183445156
Standard Deviation: 1.7644204199522617
```



Outcomes:

CO3: Apply advanced analytics to predict and optimize decision making process

Conclusion:

In this experiment, PetalLengthCm was considered as a continuous random variable and its probability distribution was analyzed using histogram and KDE plot. The mean, variance and standard deviation were calculated to understand the central tendency and spread of the data. The distribution helped in interpreting how petal length values are distributed in the Iris databset.

Grade: AA / AB / BB / BC / CC / CD /DD

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Signature of faculty in-charge with date

References:

1. G. Nandi and R. K. Sharma, Data Science: Fundamentals and Practical Approaches. New Delhi, India:BPB Publications, 2022.
2. R. E. Walpole, R. H. Myers, S. L. Myers, and K. Ye, Probability and Statistics for Engineers and Scientists, 9th ed. Boston, MA, USA: Pearson, 2010.
3. Motwani, Data Analytics Using R. Hoboken, NJ, USA: Wiley.
4. T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed. New York, NY, USA: Springer, 2014.